

O & Soergel, lec 7: Basics on category $\mathcal{O}$, III

1) Translation functors

2) Reflection functors

0) Recap

Let A be a local $S(\mathfrak{h})$ -algebra. To simplify the exposition we assume that $\chi \in \mathfrak{h}^* \otimes \mathfrak{m}$ (so we deal w. deformed $\mathcal{O}_0$ & the equivalence classes $\vec{\lambda}$ for $\sim_\chi$ are the $(W, \cdot)$ -orbits in Λ).

In Lec 6 we introduced the projective functors $\rho_{\vec{\lambda}, \vec{\lambda}'}(V \otimes L_{\vec{\lambda}}) : \mathcal{O}_A^{\vec{\lambda}} \rightarrow \mathcal{O}_A^{\vec{\lambda}'}$ admitting biadjoints $\rho_{\vec{\lambda}, \vec{\lambda}'}(V^* \otimes L_{\vec{\lambda}'})$

We've got some information on $M = \rho_{\vec{\lambda}, \vec{\lambda}'}(V \otimes \Delta_A(\lambda))$ w.

$\lambda \in \vec{\lambda}$:

Thm: M admits a filtration $\{0\} = M_0 \subsetneq M_1 \subsetneq \dots \subsetneq M_r = M$
w. $M_i / M_{i-1} \cong \Delta_A(\lambda + \mu_i)$, where $\mu_1, \dots, \mu_r$ are all weights of V s.t. $\lambda + \mu_i \in \vec{\lambda}'$ (counted w. multiplicity $\dim V_{\mu_i}$)
& ordered in such a way that $\mu_i \leq \mu_j \Rightarrow i \geq j$

1) Translation functors.

Our goal in this section is to examine special cases of Thm, where the combinatorics is easy to understand. Our setting as follows: let $\lambda_1, \lambda_2 \in \Lambda$ be s.t.

- $\lambda_1 + \rho$ is dominant
- $\mu = \lambda_2 - \lambda_1$ is dominant $\Rightarrow \lambda_2 + \rho$ is dominant.

Example: Let I denote the labelling set for simple roots ($i \in I \leadsto \alpha_i$). For $i \in I$ we write ω_i for the fundamental weight corresponding to i ($\langle \omega_i, \alpha_j^\vee \rangle = \delta_{ij}$). For $J \subset I$ write ρ_J for $\sum_{i \in J} \omega_i$ (hence $\rho = \rho_I$). We set $\lambda_1 = -\rho_J$, take dominant $\lambda_2 \Rightarrow \mu = \rho_J + \lambda_2$ is dominant.

We take $\vec{\omega}_i := W \cdot \lambda_i$, $V = L(\mu)$ (fin. dim. irrep). We write $\mathcal{T}_{\lambda_2 \rightarrow \lambda_1} := p_{\vec{\omega}_1} (V \otimes i_{\vec{\omega}_2}^*) : \mathcal{O}_A^{\vec{\omega}_2} \rightarrow \mathcal{O}_A^{\vec{\omega}_1}$
 $\mathcal{T}_{\lambda_1 \rightarrow \lambda_2} := p_{\vec{\omega}_2} (V \otimes i_{\vec{\omega}_1}) : \mathcal{O}_A^{\vec{\omega}_1} \rightarrow \mathcal{O}_A^{\vec{\omega}_2}$

By the construction, $\mathcal{T}_{\lambda_1 \rightarrow \lambda_2}$ is biadjoint to $\mathcal{T}_{\lambda_2 \rightarrow \lambda_1}$.

We note that $W_\ell := \text{Stab}_W(\lambda_\ell + \rho)$ ($\ell=1,2$) is generated by the simple reflections s_j w. $\langle \lambda_\ell + \rho, \alpha_j^\vee \rangle = 0$. In particular,

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$W_2 \subset W_1$. Choose representatives $u_1, \dots, u_r$ of the cosets in W_1/W_2 ordered so that $\ell(u_k) < \ell(u_j) \Rightarrow k > j$.

Thm: 1) $\mathcal{T}_{\lambda_2 \rightarrow \lambda_1} \Delta_A(w \cdot \lambda_2) \cong \Delta_A(w \cdot \lambda_1) \quad \forall w \in W$

2) Suppose w is shortest in wW_1 . Then the object

$M = \mathcal{T}_{\lambda_1 \rightarrow \lambda_2} \Delta_A(w \cdot \lambda_1)$ admits a filtration:

$$\{0\} = M_0 \subsetneq M_1 \subsetneq \dots \subsetneq M_r = M \quad \text{w.} \quad M_j/M_{j-1} \cong \Delta_A(w u_j \cdot \lambda_2)$$

2') In particular, if $W_2 = W_1$, then $\mathcal{T}_{\lambda_1 \rightarrow \lambda_2} \Delta_A(w \cdot \lambda_1) \cong \Delta_A(w \cdot \lambda_2)$.

Example: $\lambda_1 = -\omega_i$, λ_2 dominant, $W_1 = \{1, s_i\} \supset W_2 = \{1\}$.

Then 2) yields a SES:

$$0 \rightarrow \Delta_A(w \cdot \lambda_2) \rightarrow \mathcal{T}_{\lambda_1 \rightarrow \lambda_2} \Delta_A(w \cdot (-\omega_i)) \rightarrow \Delta_A(w s_i \cdot \lambda_2) \rightarrow 0$$

where $\ell(w) < \ell(w s_i) \Rightarrow w(\alpha_i) > 0 \Rightarrow w s_i \cdot \lambda_2 = w \cdot \lambda_2 - \underbrace{(\langle \lambda_2, \alpha_i^\vee \rangle + 1)}_{>0} w(\alpha_i)$

$< w \cdot \lambda_2$, so we have the larger highest weight in the sub,
as we should.

Proof of Thm:

1) Let ν be a weight of V . We claim that

$$\overline{3} \quad (*) \quad w \cdot \lambda_2 - \nu \in W \cdot \lambda_1 \Rightarrow \nu = w\mu$$

(*) and Thm in Sec 0 imply 1): $(V^*)_{-\lambda} = (V_{\lambda})^* \forall \lambda \in \Lambda$ & $\dim V_{w\mu} = \dim V_{\mu} = 1$ - so we get just one term in the filtration & it's $w \cdot \lambda_2 - w\mu = w \cdot (\lambda_2 - \mu) = w \cdot \lambda_1$

Let's prove (*). We have $w \cdot \lambda_2 - \lambda = u \cdot \lambda_1 \Leftrightarrow [\text{cancel } -\rho]$
 $w(\lambda_2 + \rho) - \lambda = u(\lambda_1 + \rho) \Leftrightarrow [x := w^{-1}u] \lambda_2 + \rho - x(\lambda_1 + \rho) = w^{-1}\lambda$
 $\Leftrightarrow \underbrace{(\lambda_2 + \rho) - (\lambda_1 + \rho)}_{\mu} + \underbrace{(\lambda_1 + \rho) - x(\lambda_1 + \rho)}_{\geq 0 \text{ b/c } \lambda_1 + \rho \text{ is dominant}} = w^{-1}\lambda \Rightarrow \mu \leq w^{-1}\lambda$

But μ is dominant & $w^{-1}\lambda$ is a weight of $V = L(\mu) \Rightarrow \mu \geq w^{-1}\lambda$. So $\mu = w^{-1}\lambda \Leftrightarrow \lambda = w\mu$, which is (*).

2) This proceeds similarly to 1) w. (*) replaced w.

(**) $w \cdot \lambda_1 + \lambda \in W \cdot \lambda_2 \Rightarrow \lambda = wu_j \mu$ for some $i=1, \dots, r$

(note that $u_j \cdot \lambda_1 = \lambda_1$, but $u_j \cdot \lambda_2$ are pairwise distinct $\Rightarrow u_j \mu = u_j \cdot \lambda_2 - u_j \cdot \lambda_1$ are pairwise distinct for $j=1, \dots, r$).

The proof of (**) & derivation of 2) from there are similar to part 1) & are left as **exercises** (hint: if

w, w' are shortest in their right W_2 -cosets & $w \cdot \lambda_2 < w' \cdot \lambda_2$

$\Rightarrow \ell(w') < \ell(w)$.

□

2) Reflection functors

2.1) Definition & basic properties

Let $\Xi := W \cdot \lambda$ for dominant λ & for $i \in I$, set $\lambda^i = -\omega_i$, $\Xi_i = W \cdot \lambda^i$. So we can consider the translation functors

$$\begin{aligned} \mathcal{T}_{\lambda \rightarrow \lambda_i} &: \mathcal{O}_A^\Xi \rightarrow \mathcal{O}_A^{\Xi_i} \\ \mathcal{T}_{\lambda_i \rightarrow \lambda} &: \mathcal{O}_A^{\Xi_i} \rightarrow \mathcal{O}_A^\Xi \end{aligned}$$

Here's a generalization of the reflection functors that were mentioned in Sec 2 of Lec 1.

Definition: The **reflection functor** $\mathcal{H}_i: \mathcal{O}_A^\Xi \rightarrow \mathcal{O}_A^\Xi$ is

$$\mathcal{H}_i := \mathcal{T}_{\lambda_i \rightarrow \lambda} \circ \mathcal{T}_{\lambda \rightarrow \lambda_i}$$

Here are basic properties:

Proposition: 1) $\mathcal{H}_i$ is self-adjoint (hence exact & sends projectives to projectives)

2) Let $w \in W$, then $\mathcal{H}_i \Delta_A(w \cdot \lambda)$ fits into SES:

$$0 \rightarrow \Delta_A(w \cdot \lambda) \rightarrow \bigoplus_i \Delta_A(w \cdot \lambda) \rightarrow \Delta_A(ws_i \cdot \lambda) \rightarrow 0, \text{ if } \ell(w) < \ell(ws_i)$$

$$0 \rightarrow \Delta_A(ws_i \cdot \lambda) \rightarrow \bigoplus_i \Delta_A(w \cdot \lambda) \rightarrow \Delta_A(w \cdot \lambda) \rightarrow 0, \text{ if } \ell(w) > \ell(ws_i)$$

Proof:

1) follows from the biadjunction between $T_{\lambda \rightarrow \lambda_i}$, $T_{\lambda_i \rightarrow \lambda}$.

2) follows by combining Thm & Main Example in Sec 1 $\square$

2.2) Projectives $\bigoplus_{\underline{w}} \Delta_A(\lambda)$.

Recall that we assume $\chi \in \mathcal{Y}^* \otimes \mathfrak{m}$. For a simple reflection $s = s_i$, we write $\bigoplus_s$ for $\bigoplus_i$. For a word $\underline{w} = (s_1 \dots s_k) \in S^k$ we write $\bigoplus_{\underline{w}}$ for $\bigoplus_{s_k} \dots \bigoplus_{s_1}$.

Thm: 1) $\forall \underline{w} \in S^k \Rightarrow \bigoplus_{\underline{w}} \Delta_A(\lambda)$ is projective.

2) Suppose $\underline{w}$ is a reduced expression for w . Then $\bigoplus_{\underline{w}} \Delta_A(\lambda)$ surjects onto $\Delta_A(w \cdot \lambda)$ and the kernel is filtered by $\Delta_A(u \cdot \lambda)$ w $\ell(u) < \ell(w)$.

Proof:

1) follows from 1) of Proposition in Sec 2.1 combined

6 w. Prop. in Sec 2.1 of Lec 6 (showing $\Delta_A(\lambda)$ is projective).

2) is proved by induction on ℓ . Let $\underline{w}' = (s_1, \dots, s_{\ell-1})$, the reduced expression of $w' = s_1 \dots s_{\ell-1}$, so that $w = w' s_\ell$ & $\ell(w') < \ell(w)$.

By induction (the base $\ell=0$ is vacuous), $\bigoplus_{\underline{w}'} \Delta_A(\lambda) \twoheadrightarrow \Delta_A(w' \cdot \lambda)$ w. ker filtered by $\Delta_A(u' \cdot \lambda)$ w. $\ell(u') < \ell(w')$. Then $\bigoplus_{\underline{w}} \Delta_A(\lambda) \simeq \bigoplus_{\ell} \bigoplus_{\underline{w}'} \Delta_A(\lambda)$ surjects on $\bigoplus_{\ell} \Delta_A(w' \cdot \lambda)$ w. kernel filtered by $\bigoplus_{\ell} \Delta_A(u' \cdot \lambda)$, b/c $\bigoplus_{\ell}$ is exact. By 2) of Prop'n ($\ell(w') < \ell(w)$)

$$0 \rightarrow \Delta_A(w' \cdot \lambda) \rightarrow \bigoplus_{\ell} \Delta_A(w' \cdot \lambda) \rightarrow \Delta_A(w \cdot \lambda) \rightarrow 0 \quad \&$$

$\bigoplus_{\ell} \Delta_A(u' \cdot \lambda)$ is filtered by $\Delta_A(u' \cdot \lambda)$ & $\Delta_A(u' s_\ell \cdot \lambda)$. Note that

$$\bigoplus_{\underline{w}} \Delta_A(\lambda) \twoheadrightarrow \bigoplus_{\ell} \Delta_A(w' \cdot \lambda) \twoheadrightarrow \Delta_A(w \cdot \lambda) \rightarrow 0 \quad \& \quad \ell(u' s_\ell) \leq \ell(u') + 1 \leq \ell(w') < \ell(w) \Rightarrow \ker[\bigoplus_{\underline{w}} \Delta_A(0) \twoheadrightarrow \Delta_A(w)] \text{ is filtered as claimed } \square$$

2.3) Projective generator.

Definition: Let $\mathcal{C}$ be an Λ -linear abelian category. By a **projective generator** of $\mathcal{C}$ we mean a projective $P \in \mathcal{C}$ s.t. $\forall M \in \mathcal{C} \exists n \geq 0 \& P^{\oplus n} \twoheadrightarrow M$.

We want to construct a projective generator in $\overline{\mathbb{Z}}[\mathcal{O}_A^{\Xi'}]$ for $\Xi' \in \Lambda / (W, \cdot)$

First assume $\vec{\Sigma}' = \vec{\Sigma} = W \cdot \lambda$ for dominant λ .

For each $w \in W$ pick a reduced expression $\underline{w}$. Set

$P := \bigoplus_{w \in W} \bigoplus_{\underline{w}} \Delta_A(\lambda)$, it's projective by 1) of Thm in Sec. 2.2.

Proposition 1: P is a projective generator of $\mathcal{O}_A^{\vec{\Sigma}}$.

Proof:

By 2) of Thm, $P \twoheadrightarrow \Delta(w \cdot \lambda) \nexists w \in W$. Now let $M \in \mathcal{O}_A^{\vec{\Sigma}}$. By definition, $\exists$ filtration $\{0\} = M_0 \subsetneq M_1 \subsetneq \dots \subsetneq M_r = M$ s.t. $M_i/M_{i-1} \cong \Delta(w_i \cdot \lambda)/?$. Since P is projective, an epimorphism $P \twoheadrightarrow \Delta(w_i \cdot \lambda)/?$ lifts to $P \xrightarrow{\varphi_i} M_i$. This gives $P^{\oplus r} \rightarrow M$, $(p_1, \dots, p_r) \mapsto \sum_{i=1}^r \varphi_i(p_i)$. It's surjective (exercise) $\square$

Now let $\lambda' \in \Lambda$ be s.t. $\lambda' + \rho$ be dominant & $\vec{\Sigma}' = W \cdot \lambda'$.

Choose dominant λ s.t. $\lambda - \lambda'$ is dominant. We have the projective generator $P \in \mathcal{O}_A^{W \cdot \lambda}$ as above.

Proposition 2: $T_{\lambda \rightarrow \lambda'}(P)$ is a projective generator on $\mathcal{O}_A^{\vec{\Sigma}'}$.

Proof: Note that $\mathcal{T}_{\lambda \rightarrow \lambda'}(P)$ is projective. Also

$P \rightarrow \Delta(w \cdot \lambda) \Rightarrow \mathcal{T}_{\lambda \rightarrow \lambda'}(P) \rightarrow \mathcal{T}_{\lambda \rightarrow \lambda'} \Delta(w \cdot \lambda) \simeq [1]$ of Thm in
Sec 1] $\Delta(w \cdot \lambda')$. Then we can argue as in the proof of Propo-
sition 1 □